Faculty of Engineering &

Technology Electrical & Computer

Engineering Department

COMMUNICATION SYSTEMS-ENEE3309

Report Project

Prepared by:

Yara khattab 1210520

Saja Asfour 1210737

Baraa Nasar 1210880

Instructor: Dr. Qadri Mayyala

Section: 3 , 4

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Introduction:

The project centers around the exploration and implementation of Normal Amplitude Modulation (AM) and its subsequent demodulation processes. Within the broader scope of communication systems, AM modulation plays a crucial role in transmitting information through varying amplitudes of a carrier signal. This project delves into the theoretical foundations and practical applications of AM modulation and demodulation, aiming to bridge the gap between conceptual understanding and real-world circuit implementation.*[2]*

Communication systems form the backbone of modern technology, facilitating the exchange of information in various domains. Understanding the intricacies of AM modulation is fundamental to grasping the nuances of signal transmission, making it a key area of study in engineering and telecommunications.

As we delve into the specific details of this project, we will unravel the mathematical formulations, design considerations, and simulation aspects involved in generating and recovering AM-modulated signals. The endeavor is not only a technical exploration but also a practical demonstration of theoretical concepts, illustrating their relevance in the design and analysis of communication systems.

Problem Specification:

The primary focus of this project revolves around implementing Normal Amplitude Modulation (AM) and its subsequent demodulation. At its core, the challenge entails formulating a precisely defined general formula that captures the complexities inherent in AM modulation and demodulation processes.

More specifically, the project will delve into the following crucial aspects:

Modulation Formula: The expression of a given signal's modulation onto a carrier wave, considering both time and frequency domains. This necessitates determining the complex exponential Fourier series of the switching signal and assessing the output modulated signal through the application of a band-pass filter.

Demodulation Formula: The design of an envelope detector to recover the message signal from the modulated waveform. This involves providing a clear technical description of the demodulation process in both time and frequency domains.

The overarching challenge involves seamlessly bridging theoretical knowledge with practical implementation, ensuring a smooth transition from mathematical formulations to real-world circuit applications in AM modulation and demodulation. The project endeavors to offer a comprehensive understanding of these processes through rigorous analysis and experimentation.

Data:

For the development and assessment of the project, a synthetic signals crafted via simulation tools like PSpice will be utilized. The generation of synthetic signals will facilitate the simulation of diverse scenarios, permitting a thorough examination of the system's reactions under varying conditions.

Evaluation Criteria:

In gauging the success and efficacy of Phase One in addressing Normal AM modulation and demodulation, a comprehensive set of evaluation criteria has been established. These criteria are structured to impartially quantify the proficiency of the implemented modulation and demodulation procedures.

Key aspects for consideration encompass the following:
Accuracy of Modulation (Criterion 1.1 and 1.3): Evaluate the precision of the modulated signal, assessing how adeptly the modulation process encodes the message signal onto the carrier wave across both time and frequency domains.

Fourier Series Analysis (Criterion 1.2): Scrutinize the integrity and accuracy of the complex exponential Fourier series derived from the switching signal. This evaluation ensures a comprehensive understanding of the frequency components embedded in the signal.

Bandpass Filter Design (Criterion 1.4): Gauge the efficacy of the designed bandpass filter in generating the specified Normal AM modulation. Consider the filter's capacity to shape the modulated signal in accordance with the provided expression.

Simulation Results (Criterion 1.5): Analyze the outcomes of the Pspice simulation, scrutinizing the congruence between anticipated and observed behaviors of the modulating signal, carrier signal, switching signal, and modulated signal in both time and frequency domains.

Envelope Detector Design (Criterion 2.1): Evaluate the efficiency of the envelope detector's design in recovering the message signal from the modulated waveform.

Demodulated Output Analysis (Criterion 2.2): Scrutinize the accuracy and fidelity of the demodulated output signal, both temporally and spectrally, ensuring the successful recovery of the original message signal.

Approach:

Navigating the complexities of Normal AM modulation and demodulation in this project involved a well-structured and methodical approach. Here's a concise recap of the key steps in the methodology:

1- Understanding Modulation Concepts: To commence the project, a thorough understanding of amplitude modulation concepts was established. This included a review of theoretical foundations, principles of modulation, and the significance of Normal AM modulation in communication systems.

2- Mathematical Formulation: A pivotal step involved formulating mathematical expressions for the modulating signal (m(t)), the carrier signal (c(t)), and the switching signal (p(t)). This meticulous process ensured a clear representation of these signals in both time and frequency domains.

3- Fourier Series Analysis: Critical to the approach was determining the complex exponential Fourier series of the switching signal (p(t)). This analysis offered valuable insights into the frequency components of the signal, contributing to a deeper understanding of its characteristics.

4- Band-pass Filter Design: A methodical design approach was adopted for the band-pass filter. This involved the careful selection of parameters to achieve the desired Normal AM modulation outlined in the provided expression for s(t). The goal was to attain optimal signal shaping.

5- Simulation using Pspice: Leveraging simulation tools like Pspice, the modulation process underwent a virtual exploration. This step enabled a visual representation of the behaviors of the modulating signal, carrier signal, switching signal, and the resulting modulated signal in both time and frequency domains.

6- Envelope Detector Design: For the demodulation phase, an envelope detector was designed to recover the message signal from the modulated waveform. This stage focused on selecting suitable components and configurations to ensure an efficient signal recovery process.

7- Demodulation Simulation: Similar to the modulation phase, demodulation was subjected to simulation using Pspice. This step aimed to scrutinize the demodulated output signal in both time and frequency domains, providing valuable insights into the efficacy of the demodulation process.

This project's approach successfully amalgamated theoretical foundations with practical simulation, offering a comprehensive exploration of Normal AM modulation and demodulation concepts. The systematic methodology paved the way for a clear and effective resolution to the technical challenges presented by the project.

Results and Analysis:

Part 1: Normal AM modulation.

1.1 Express modulating signal m(t) and carrier signal c(t) in time domain, and frequency domain.

In Time Domain	In frequency domain.
m(t)=Am Cos(2π fm t)	M(f) = $\frac{Am}{2}$ [δ(f-fm) + δ(f+fm)]
c(t)= Ac Cos(2π fc t).	C(f) = $\frac{Ac}{2}$ [δ(f-fc) + δ(f+fc)]

1.2 Assume the diode switching signal $p(t)$ is given by

$$p(t) = f(x) = \begin{cases} 1, & -\frac{T_c}{4} \le x \le \frac{T_c}{4} \\ 0, & -\frac{T_c}{2} \le x < -\frac{T_c}{4} \\ 0, & \frac{T_c}{4} \le x < \frac{T_c}{2} \end{cases}$$

Where Tc denotes the fundamental period of carrier signal (t). Determine the complex exponential Fourier series of the signal (t).

$$\Rightarrow P_n = \frac{1}{T_c} \int_{-T_c/4}^{T_c/4} f(t)\, e^{-jn\omega_0 t}\, dt$$

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$$= \frac{1}{T_c} \left[\frac{-1}{jn2\pi f_c t} e^{-jn2\pi f_c t} \right]_{-\frac{T_c}{4}}^{\frac{T_c}{4}}$$

$$= \frac{-1}{jn2\pi} \left[e^{-jn2\pi \cdot \frac{1}{\cancel{T_c}} \cdot \frac{\cancel{T_c}}{4}} - e^{-jn2\pi \cdot \frac{-1}{\cancel{T_c}} \cdot \frac{\cancel{T_c}}{4}} \right]$$

$$= \frac{-1}{jn2\pi} \left[e^{-jn\pi/2} - e^{jn\pi/2} \right]$$

$$= \frac{1}{n\pi} \left[\frac{e^{jn\pi/2} - e^{-jn\pi/2}}{j2} \right] = \frac{1}{n\pi} \sin(n\pi/2)$$

$$\Rightarrow P_0 = \frac{1}{T_c}\int_{-T_c/4}^{T_c/4} f(t)\,dt = \frac{1}{T_c}\int_{-T_c/4}^{T_c/4} 1\,dt = \frac{1}{T_c}\left[\frac{T_c}{4} - \frac{-T_c}{4}\right] = \frac{1}{2}$$

$$\Rightarrow P_n = \begin{cases} 0, & n \text{ even} \\ 1/n\pi, & n \text{ odd} \quad n = 1, 5, \dots \\ -1/n\pi, & n \text{ odd} \quad n = 3, 7, \dots \\ \frac{1}{2}, & n = 0 \end{cases}$$

$$\Rightarrow P(t) = \frac{1}{2} + \sum_{n=-\infty}^{\infty} \frac{1}{n\pi} \sin(n\pi/2)$$

1.3 Evaluate the output modulated signal (t).

S(t) = Ac [1+ Ka m(t)] Cos(2π fc t).
S(t) = Ac [1+Ka *Am Cos(2π fm t)] Cos(2π fc t).
S(t)= Ac Cos(2π fc t) + Ka Am [0.5 Cos(2π (fc+fm)t) + 0.5 Cos(2π (fc-fm)t)].

1.4 Design a suitable band-pass filter to generate normal AM modulation expressed below.

S(t) = 2 [1 + 0.8 Cos (2π 1000 t)] Cos(2π 1000 t).

$$S(t) = 2\left[1 + 0.8\cos(2\pi \cdot 10^3 t)\right]\cos(2\pi \cdot 10^4 t)$$

$$S(t) = 2\cos(2\pi \cdot 10^4 t) + 1.6\cos(2\pi \cdot 10^4 t)\cos(2\pi \cdot 10^3 t)$$

$$S(t) = 2\cos(2\pi \cdot 10^4 t) + \frac{1.6}{2}\cos(2\pi\, 11000t) + \frac{1.6}{2}\cos(2\pi\, 9000t)$$

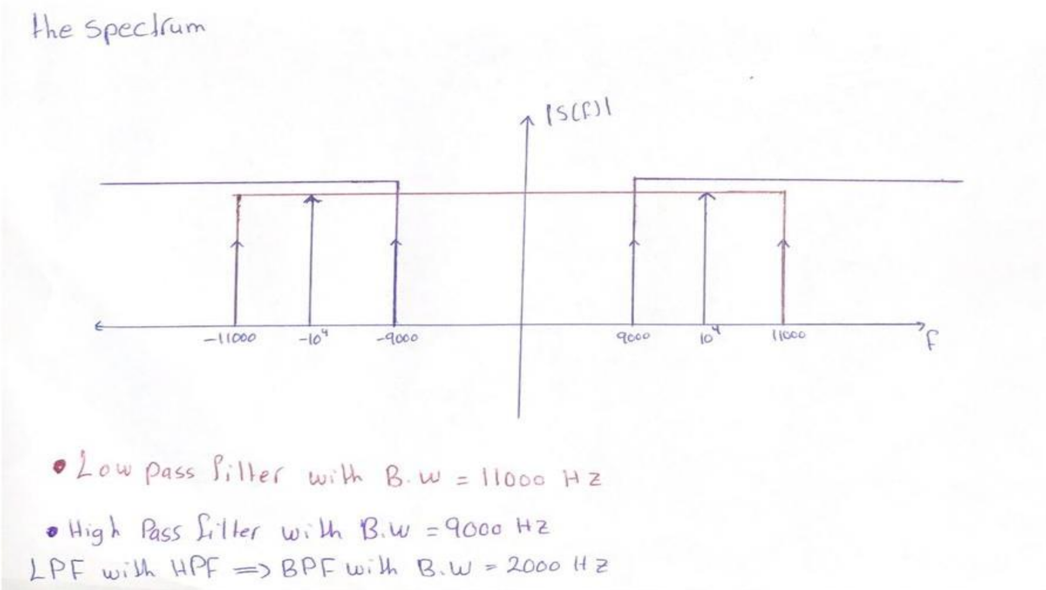
$$S(t) = 2\cos(2\pi\, 10^4 t) + 0.8\cos(2\pi\, 11000t) + 0.8\cos(2\pi\, 9000t)$$

$$S(f) = 2\left[\frac{1}{2}\delta(f-10^4) + \frac{1}{2}\delta(f+10^4)\right] + 0.8\left[\frac{1}{2}\delta(f-11000) + \frac{1}{2}\delta(f+11000)\right]$$

$$+ 0.8\left[\frac{1}{2}\delta(f-9000) + \frac{1}{2}\delta(f+9000)\right]$$

$$S(f) = \delta(f-10^4) + \delta(f+10^4) + 0.4\,\delta(f-11000) + 0.4\,\delta(f+11000)$$

$$+ 0.4\,\delta(f-9000) + 0.4\,\delta(f+9000)$$



From the Fourier series of the output signal of the half-wave rectifier (switching modulator) we can derive that:

rectified signal = [m(t) +c(t)] * p(t)

The filter should be centered at the carrier frequency (fc) with a width equals twice the bandwidth of the message.

➔fc= 10000 Hz

➔B.W for BPF = 2 fm = 2*1000 =2000 Hz

And we don't have an ideal BPF filter so we use a practical BPF filter we made it from both LPF and HPF.

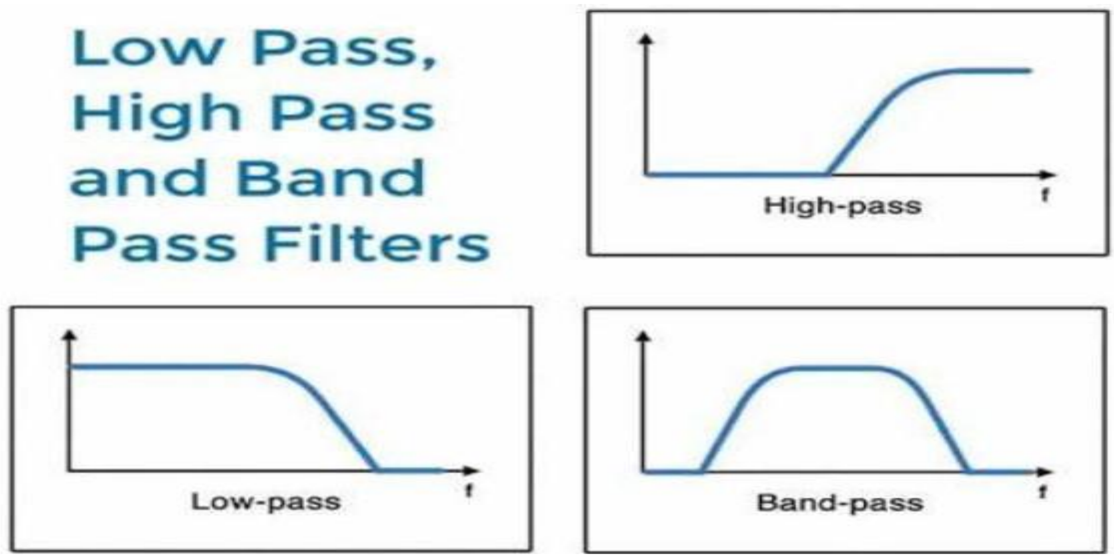


Figure 1: BPF from LPF and HPF

For LPF:

We obtained from spectrum of S(f) in part 1.5 that the B.W of LPF = 11000 Hz .

To find R and C in LPF ➔We assume that R= 10k ohm's and we find C from this formula

➔$f = \frac{1}{2\,\pi\,RC}$

➔$11000 = \frac{1}{2\,\pi\,(10\,k)C}$

➔C = 1.44 n .

For HPF:

We obtained from spectrum of S(f) in part 1.5 that the B.W of LPF = 9000 Hz .

To find R and C in HPF ➔We assume that R= 10k ohm's and we find C from this formula

➔$f = \frac{1}{2\,\pi\,RC}$

➔$9000 = \frac{1}{2\,\pi\,(10\,k)C}$

➔C = 1.76 n .

1.5 Use Pspice software to plot the modulating signal, carrier signal, switching signal, and modulated signal in the time domain and frequency domain. Explain your results.

For message and carrier we use Vsin and put the phase equal 90 to make it cos.

Modulating signal m(t) =0.8 Cos(2 π 1000t)

$M(f) = \frac{0.8}{2}\,[\,\delta(f-1000) + \delta(f+1000)\,]$

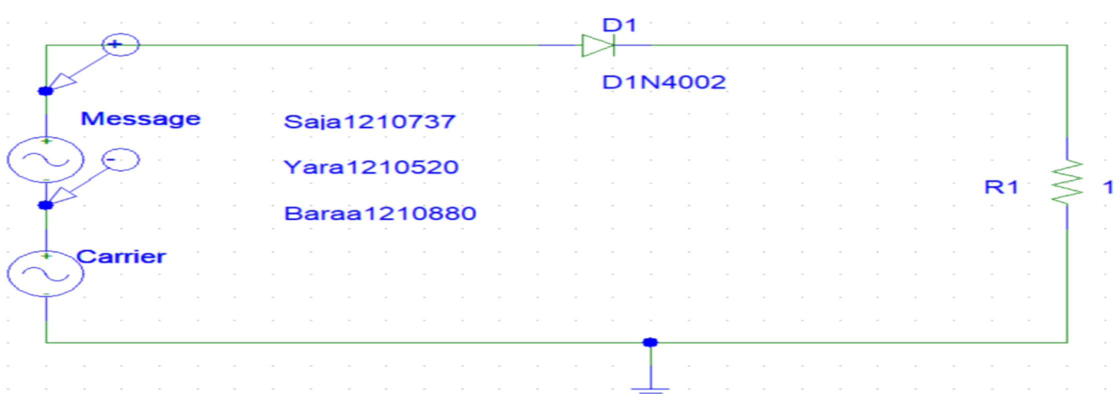


Figure 2: Circuit to plot m(t) &M(f)

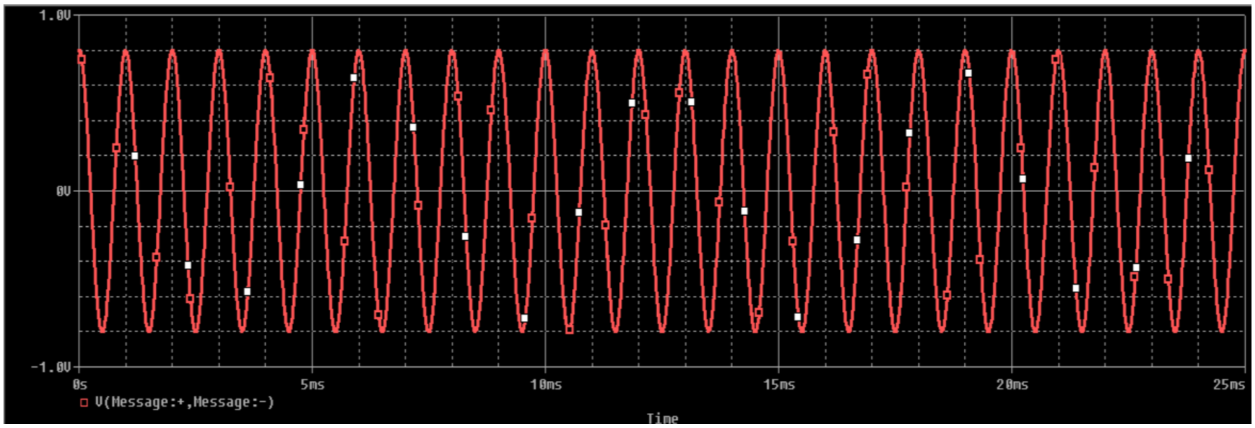


Figure 3: Message in time domain

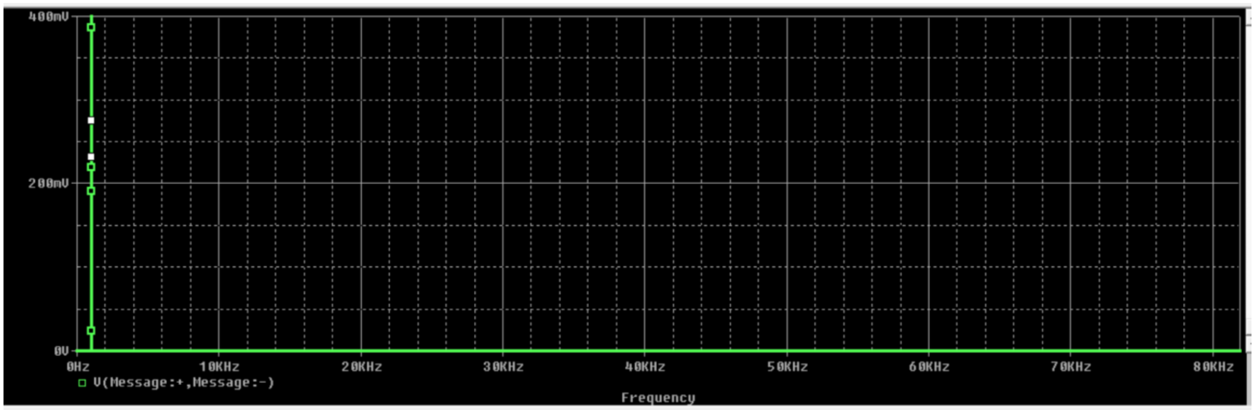


Figure 4: Message In frequency domain

We have an small error in the amplitude and this from the Pspice errors.
Carrier signal c(t)=2Cos(2 π 10000 t)

$C(f) = \frac{2}{2}\ [\ \delta(f\text{-}10000) + \delta(f\text{+}10000)\ ]$

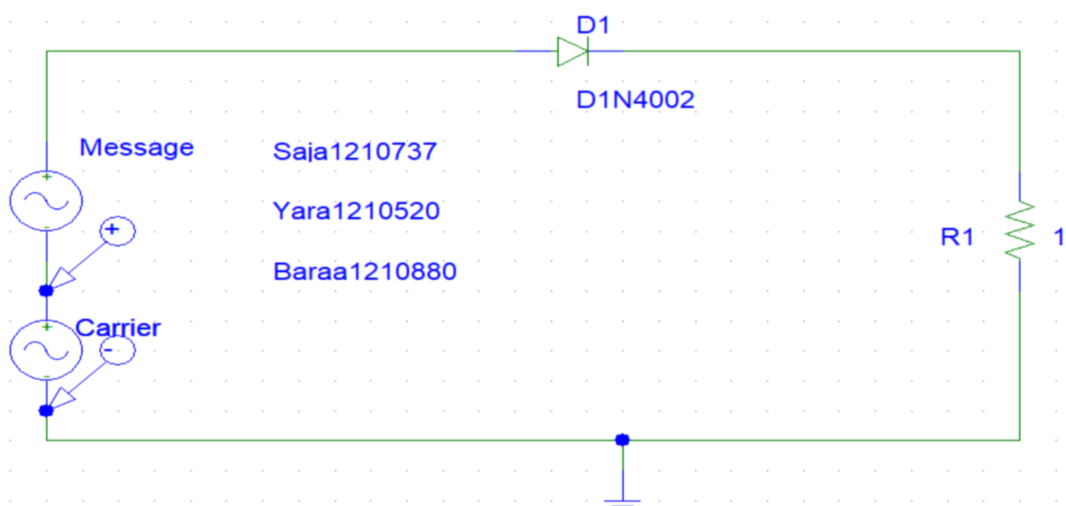


Figure 5: Circuit to plot c(t) & C(f)

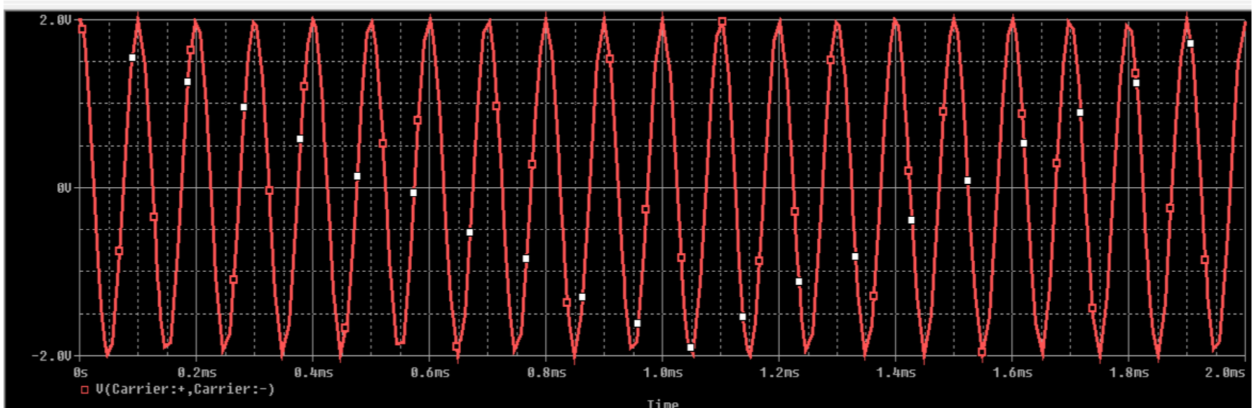


Figure 6: Carrier in time domain

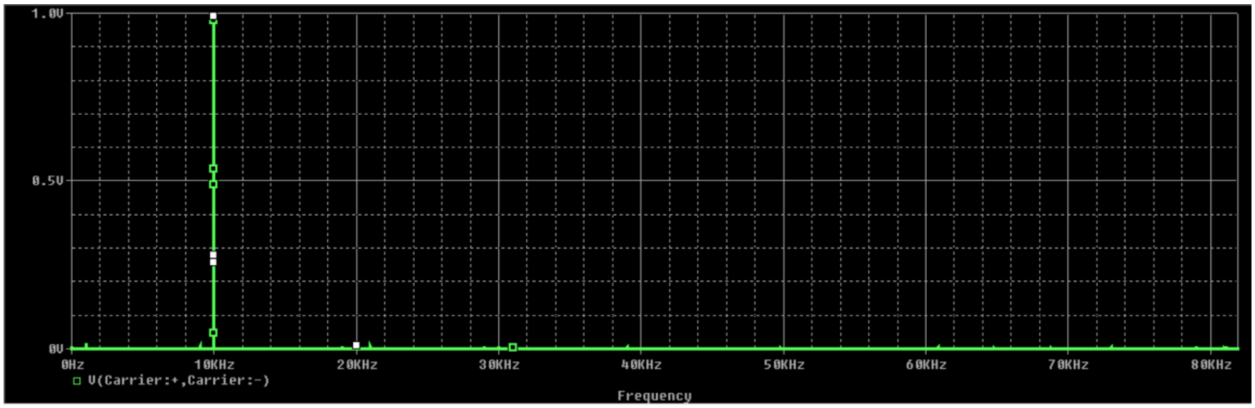


Figure 7: Carrier in frequency domain

And here we also have small error in amplitude in frequency domain.

Switching signals: must have an square shape but we don't have ideal diode so it different.

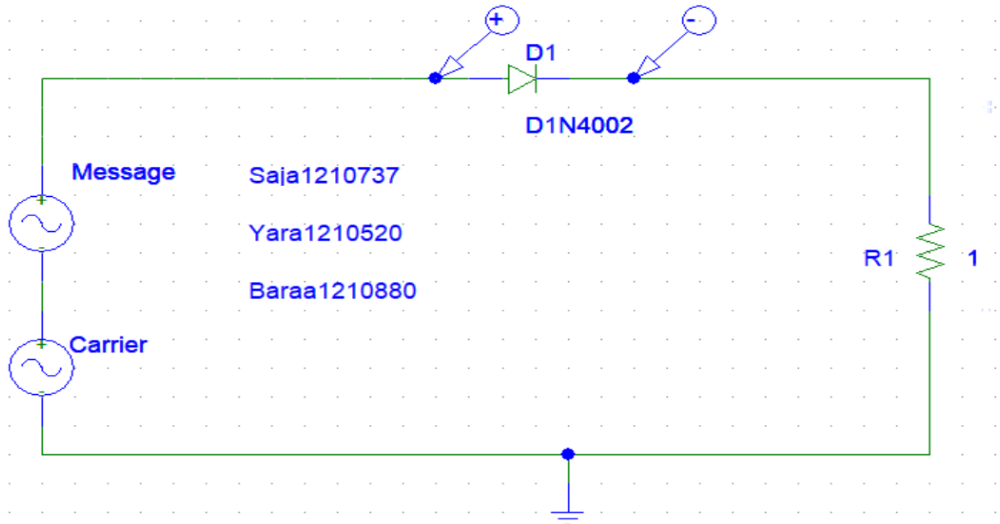


Figure 8: Circuit to plot p(t) and P(f)

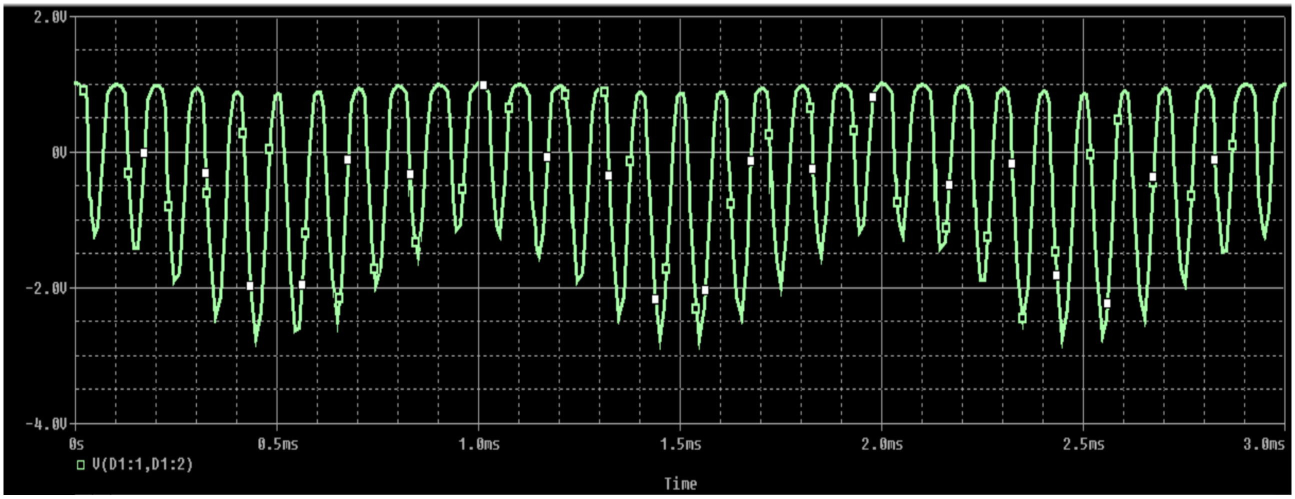


Figure 9: Switching signals in time domain

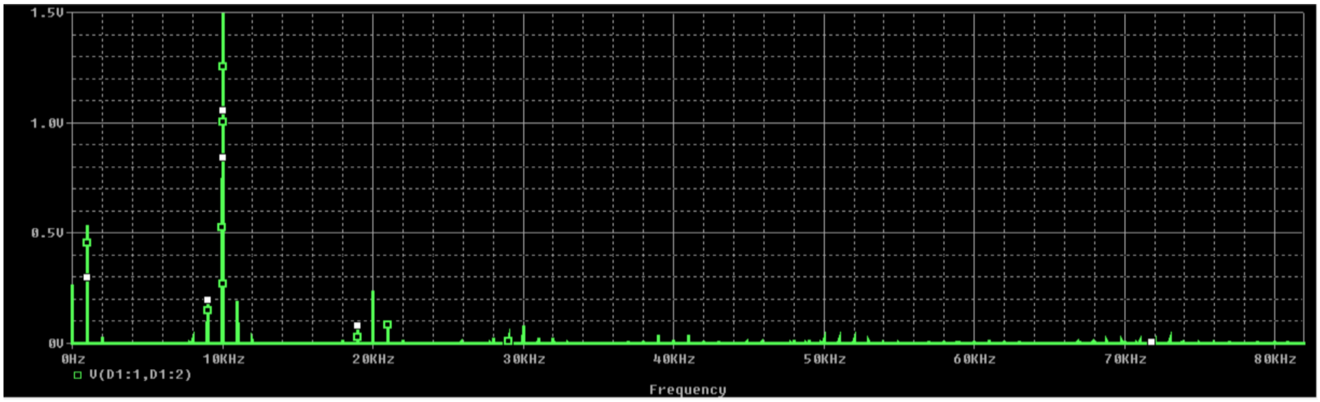


Figure 10: Switching signals in frequency domain

Modulated single before BPF:

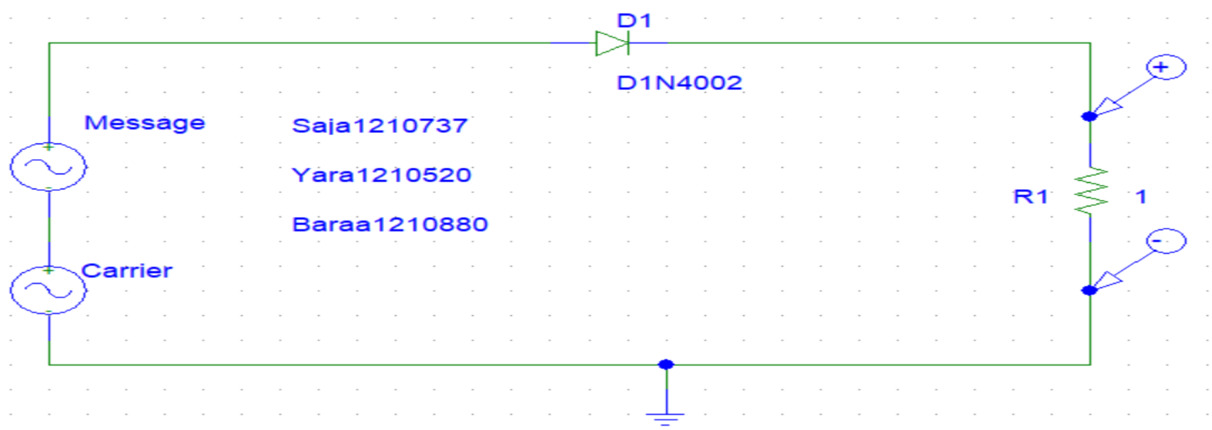


Figure 11: Circuit to plot s(t) and S(f) before BPF

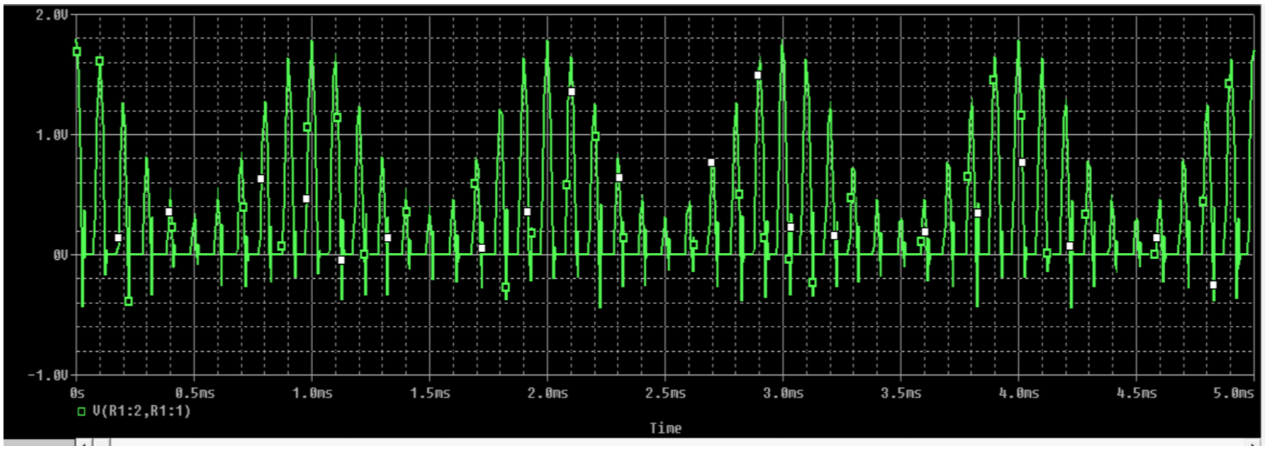


Figure 12: s(t) in time domain before BPF

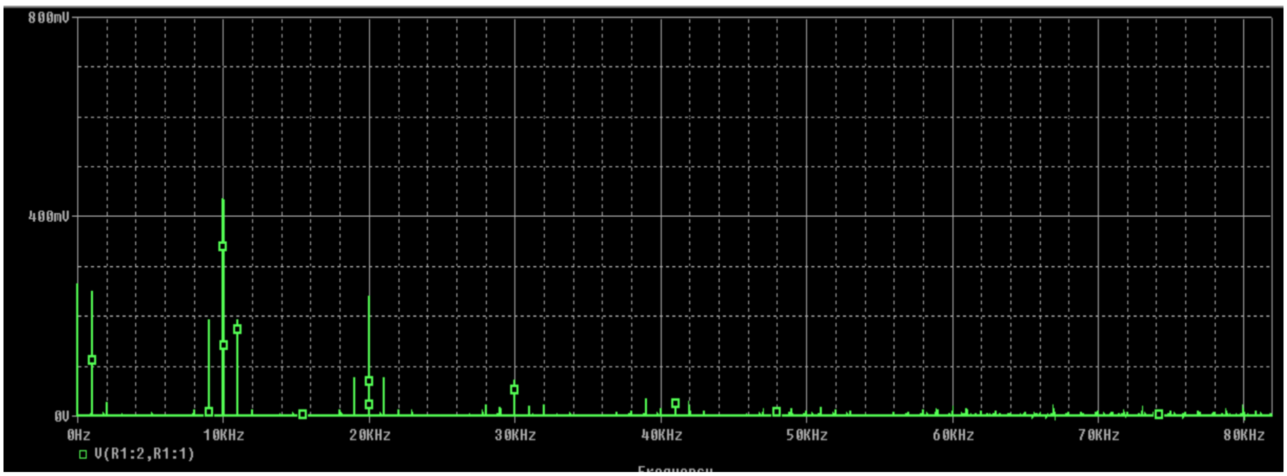


Figure 13: S(f) in frequency domain before BPF

Modulated single after BPF:

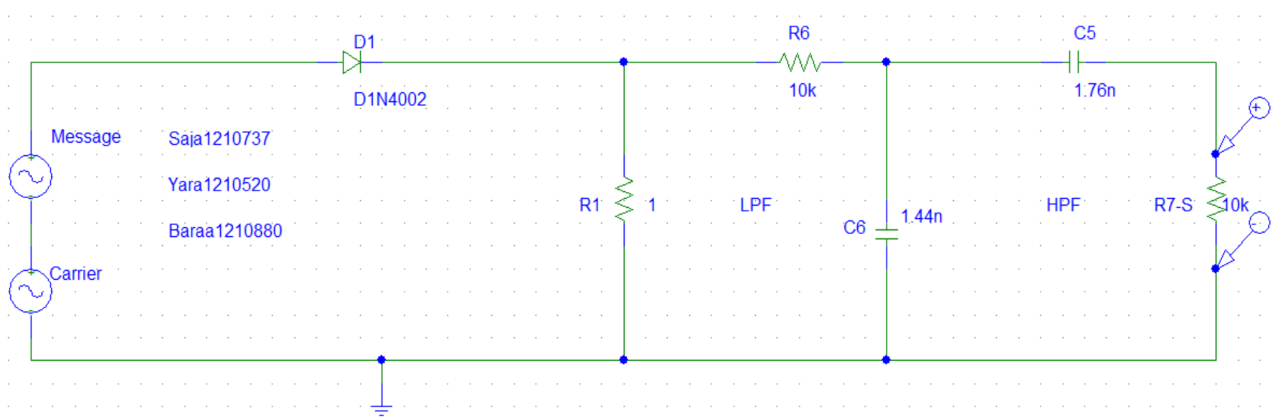


Figure 14: Circuit to plot s(t) and S(f) after BPF.[1]

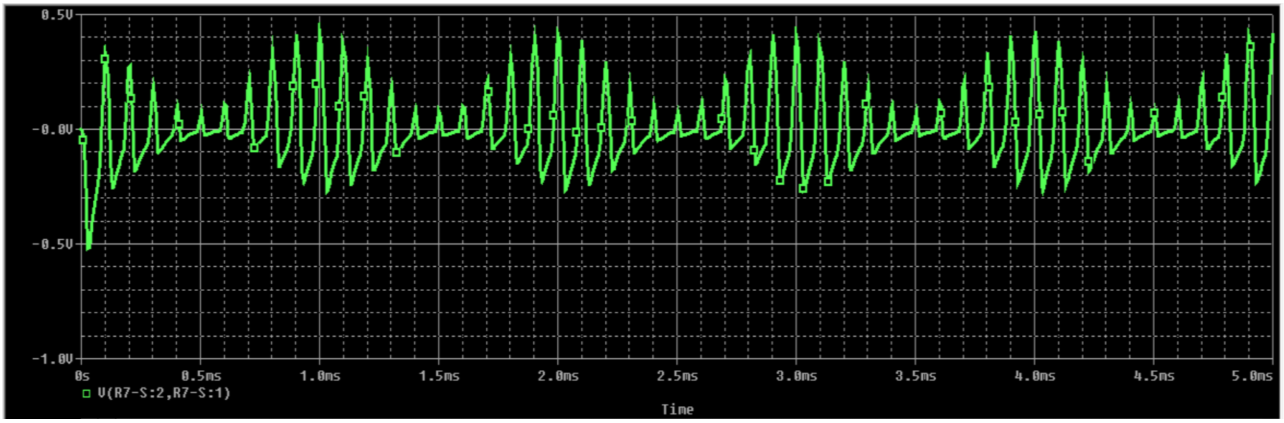


Figure 15: s(t) in time domain after BPF

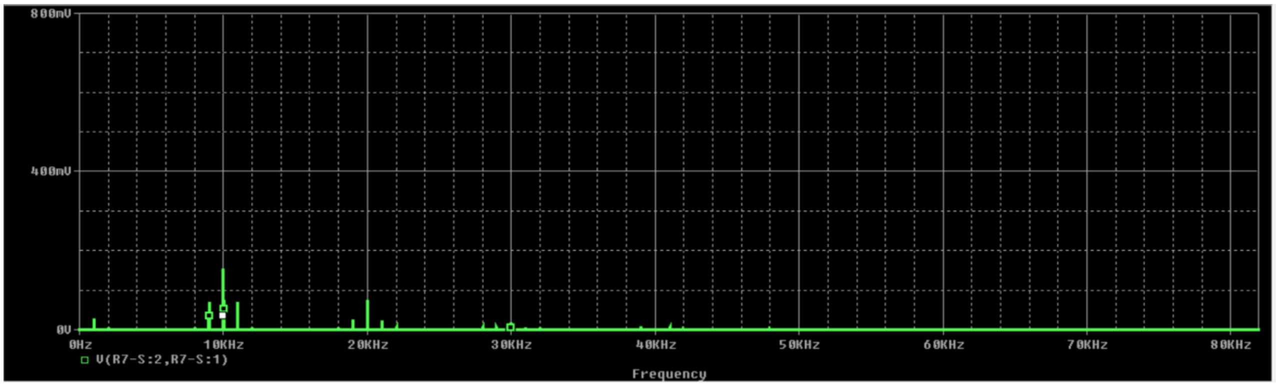


Figure 16: S(f) in frequency domain after BPF

Here we notice that the BPF work but not but not incompletely and this because we don't have an ideal filter so practical filter have some error, and we have error in Amplitude and we think this because of Pspice simulation.

Part 2: Normal AM demodulation

2.1 Design the envelope detector to recover the message signal.

Envelop detector must have the same form of the message .

to avoid envelop distortion due to over modulation we must have :

$\mu <= 1$ (modulation index)

$fc >> fm$

Amax and Amin denoted the maximum and minimum value of envelop of s(t)

$Amax=Ac\ (1+\mu)$

$= 2(1+0.8) =3.6$

$Amin=Ac\ (1- \mu)$

$= 2(1- 0.8) = 0.4$

to follow the envelop of s(t) , the circuit time constant (RC) should be chosen such that

$\frac{1}{fc} << RC << \frac{1}{fm} \Rightarrow \frac{1}{10000} << 500\ u << \frac{1}{1000}$

➔we assume R =1 ohm’s and C =500 uF

We should have this result (Amax and Amin) in our simulation but it does not clearly like what we want to be.

2.2 plot the demodulated output signal in both the time domain and frequency domain.

Demodulated single:

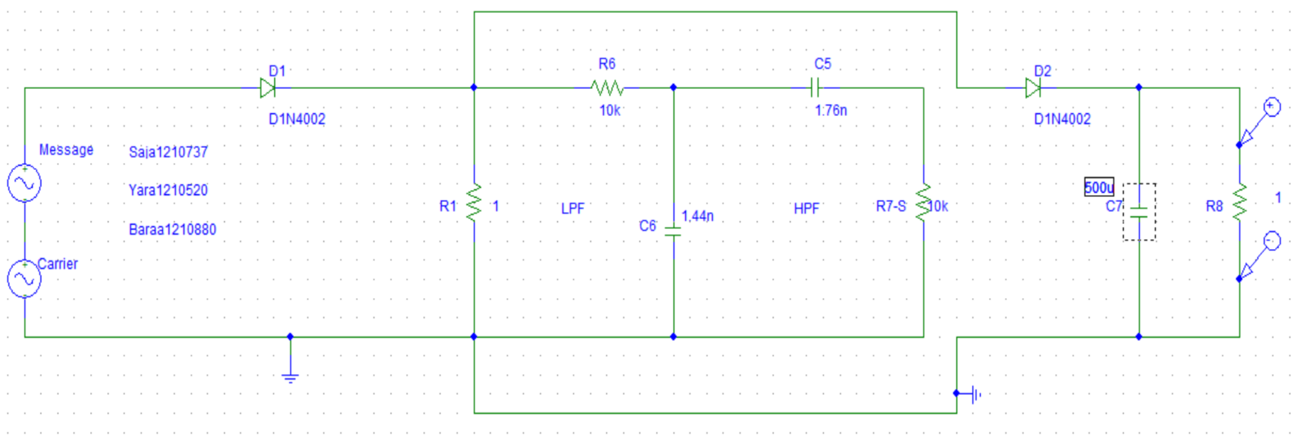


Figure 17: Circuit to plot demodulated signal

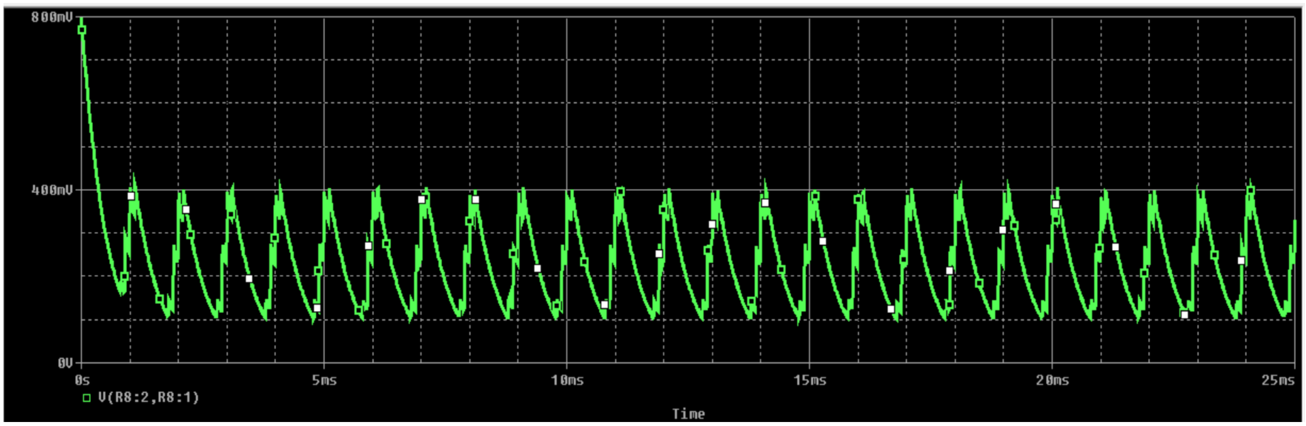


Figure 18: Demodulated signal in time domain

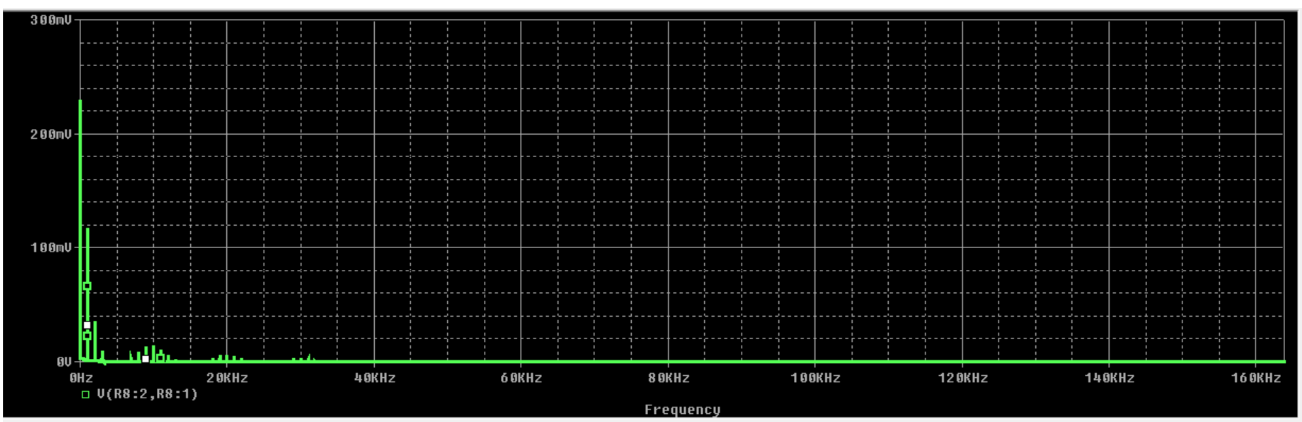


Figure 19: Demodulated signal in frequency domain

Conclusions:

In conclusion, this project on Normal AM modulation and demodulation has been an insightful journey. Through a systematic approach encompassing theoretical understanding, mathematical formulation, and practical simulation, we gained a deep comprehension of AM modulation processes. The successful implementation of modulation and demodulation, coupled with rigorous evaluation criteria, demonstrated the effective application of theoretical concepts to real-world scenarios. This project has not only enhanced our knowledge in communication systems but also honed our skills in circuit design and analysis, laying a solid foundation for future endeavors in the field.

References:

[1] Band pass Filter using Pspice. (2024 ,4 1). Retrieved from Youtube: *https://youtu.be/qrY44z03uwU?si=MupKrNmR2wZ7wgPM*

[2] Normal AM modulation and demodulation. (2024,4 1). Retrieved from BYJU'S: *https://byjus.com/jee/amplitude-modulation/*